# Automated Trading System

Market data, news sentiment and rule-based trading signals

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## 1 Project overview

This project is an automated trading system written in Python. It monitors 50 US stocks and makes a BUY, SELL or HOLD decision for each of them once per trading day. The decision is based on technical indicators and the sentiment of recent news articles.

When the signal changes, the system can place an order through the Alpaca paper-trading API. Each trade has a stop-loss and take-profit level. Signals and trades are also saved to log files, while important updates are sent to Telegram.

The main aim was to build a system that handles the whole process automatically: collecting data, calculating indicators, evaluating signals, applying risk rules and sending notifications. The system uses simulated money and does not place real trades.

## 2 System structure

The system is divided into several Python modules. Each module is responsible for a separate part of the process. Most settings, such as indicator weights and risk limits, are stored in one configuration file.

| No. | Module | Purpose |
|---|---|---|
| 1 | `main.py` | Starts the daily scan, checks whether signals have changed, saves logs and schedules the next run. |
| 2 | `main.py` / `yfinance` | Downloads six months of daily OHLCV data for each stock. Prices are adjusted for splits and dividends. |
| 3 | `indicators.py` | Calculates RSI, MACD, Bollinger Bands, ATR, Stochastic, Williams %R, CCI, OBV, moving averages and average volume. |
| 4 | `sentiment.py` | Collects recent news, calculates sentiment with VADER and combines it with price data. |
| 5 | `signals.py` | Calculates BUY and SELL scores and selects the final signal. |
| 6 | `ai_scorer.py` | Checks whether several independent conditions support the same direction. This is a local rule-based component. |
| 7 | `broker.py` | Calculates position size, checks risk limits and sends orders to Alpaca. |
| 8 | `notifier.py` | Sends signal, trade, error and daily-summary messages to Telegram. |

The scan is scheduled for weekdays at 21:30 UTC, after the US market close. A `-now` argument can also be used to start it manually.

## 3 Data sources

The system uses two main data sources:

- **Stock prices.** Daily OHLCV data is downloaded from Yahoo Finance using the `yfinance` library. The data covers the previous six months and is adjusted for splits and dividends.
- **News articles.** NewsAPI is used to find English-language articles published during the previous seven days. The search uses the stock ticker and the word "stock", with up to 100 articles collected for each ticker.

The stock list contains 50 companies from four sectors: technology (18), healthcare and biotechnology (12), energy (10) and finance (10).

## 4 Technical indicators and scoring

Each indicator can contribute to either the BUY or SELL score. If its conditions are not met, it does not add points. I gave higher weights to indicators related to trend and momentum, while oscillators have slightly lower weights.

| **Indicator** | **BUY condition** | **SELL condition** | **Weight** |
|---|---|---|---|
| RSI (14) | RSI below 35 | RSI above 70 | 1.5 |
| MACD (12, 26, 9) | MACD line above signal line | MACD line below signal line | 1.5 |
| Bollinger Bands (20, $2\sigma$) | Close below lower band | Close above upper band | 1.0 |
| SMA crossover (20 / 50) | SMA 20 above SMA 50 | SMA 20 below SMA 50 | 1.5 |
| EMA crossover (9 / 21) | EMA 9 above EMA 21 | EMA 9 below EMA 21 | 1.0 |
| Stochastic (14, 3) | %K below 20 and above %D | %K above 80 and below %D | 1.0 |
| Williams %R (14) | Below $-80$ | Above $-20$ | 0.8 |
| CCI (20) | Below $-100$ | Above 100 | 0.8 |
| Volume | Above 1.5 times the 20-day average, when BUY score leads | Same volume condition, when SELL score leads | 1.0 |
| News sentiment | Smoothed VADER score above $+0.10$ | Smoothed VADER score below $-0.10$ | 1.5 |
| Consensus layer | BUY verdict with confidence $\geq 0.40$ | SELL verdict with confidence $\geq 0.40$ | 2.5 |

The maximum possible score on either side is 14.1. The system does not make a decision based on one indicator alone, since the final score depends on several conditions.

## 5 News sentiment

For each article, the title and description are analysed using VADER. The analyser returns a compound sentiment score between $-1$ (very negative) and $+1$ (very positive).

Scores are averaged for each day and then smoothed using a three-day rolling mean. The resulting daily sentiment value is joined to the price data by date. If no news is available for a particular day, the sentiment value is set to 0.

## 6 Consensus layer

The consensus layer is a local rule-based component. It checks five conditions using the latest available price data and counts how many of them support the same direction:

- **Trend:** the closing price is more than 0.5% above or below the 50-day moving average.
- **Momentum:** the MACD histogram is positive and rising, or negative and falling.
- **RSI zone:** values from 30 to 50 are treated as a bullish recovery, while values from 55 to 75 are treated as bearish.
- **Short-term price change:** the price has changed by more than +2% or −2% over the last five days.
- **Bollinger Bands:** the closing price is in the lowest or highest quarter of the band range.

Confidence is calculated as the number of conditions supporting a direction divided by five. At least two conditions must agree for the layer to return BUY or SELL, which means confidence must be at least 0.40. Otherwise, the result is HOLD.

If the confidence threshold is reached, the consensus layer adds 2.5 points to the corresponding score.

## 7 Final trading decision

The final decision is based on the BUY and SELL scores:

- BUY is selected when the BUY score is at least 4.0 and is higher than the SELL score.
- SELL is selected when the SELL score is at least 4.0 and is higher than the BUY score.
- In all other cases, the decision is HOLD.

An order is sent only if the new signal is different from the signal saved during the previous scan. This prevents the system from placing the same order every day when the signal has not changed.

## 8 Order execution and risk management

Orders are sent to Alpaca using its paper-trading API. The system uses market orders. BUY orders are submitted as bracket orders, which means that stop-loss and take-profit levels are attached when the position is opened.

| Parameter | Value | Purpose |
|---|---|---|
| Position size | Up to 4% of portfolio value, limited to 95% of buying power | Limits exposure to one stock. |
| Stop-loss | 5% below entry price | Limits the loss on an individual trade. |
| Take-profit | 15% above entry price | Sets a 3:1 reward-to-risk ratio. |
| Maximum open positions | 12 | Keeps part of the portfolio in cash. |
| Duplicate protection | No second BUY if a position already exists | Prevents multiple positions in the same stock. |
| SELL handling | Closes the full position; skipped if no position is held | Prevents short selling. |
| Trading switch | `TRADING_ENABLED` flag | When disabled, the system only records signals and simulates decisions. |

# 9 Automation and monitoring

APScheduler runs the scan once per weekday at 21:30 UTC. The system can also be started manually with the `-now` argument.

Each scan saves the signal, scores, indicator values, sentiment and consensus information to a CSV file. Executed trades are recorded separately, including the quantity, execution price, stop-loss, take-profit and order ID.

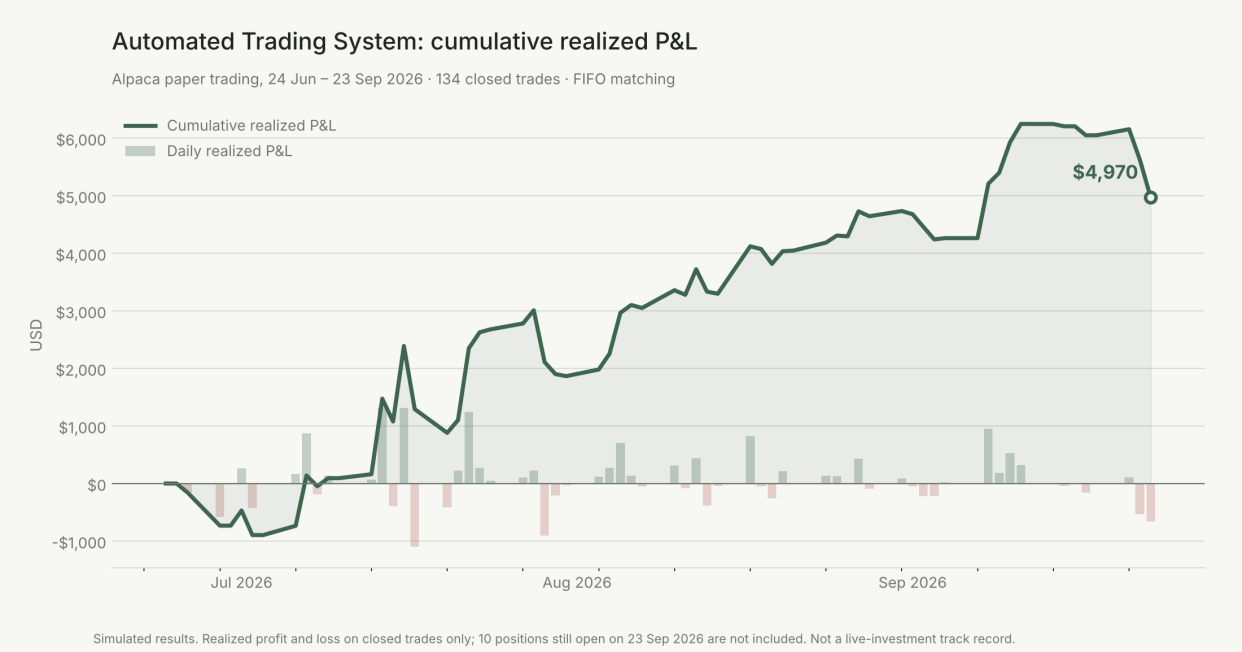
The last signal for each ticker is stored in a JSON file. This information is used during the next scan to check whether a signal has changed.

Telegram notifications are sent when a new signal appears, when a trade is executed, if an error occurs, and when the daily scan is finished. The daily summary includes the results for all tickers.

An error while processing one ticker does not stop the whole scan. The error is logged and reported, and the system continues with the remaining stocks. API requests also use timeouts.

# 10 Results

The system was tested in an Alpaca paper-trading account for three months. During this period, the approximate return was 2% per month.




These results are based on simulated trades and a relatively short testing period. They should not be treated as proof of long-term profitability. The system has not been evaluated using a formal out-of-sample backtest, and the results do not guarantee future performance.

## 11 Limitations and possible improvements

There are several limitations that should be considered when evaluating the results:

- **Short testing period.** Three months is not enough to draw reliable conclusions about long-term performance or to separate the effect of the strategy from general market conditions.
- **Manually selected parameters.** Indicator thresholds and weights were selected manually. They have not been optimised or tested on a separate dataset.
- **News sentiment.** The news search covers only the previous seven days. VADER is a general-purpose sentiment analyser and is not specifically designed for financial news.
- **Execution differences.** Signals are calculated using daily closing prices, while orders are placed after the market close. In real trading, overnight price changes, spreads and fees could lead to different execution prices. Paper trading does not fully represent these conditions.